

Richitha Rekula

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Summary

Master of Science in Data Science, Analytics and Engineering candidate at Arizona State University, targeting software engineering, applied AI, and AI/ML engineering roles. Builds and tests machine learning and generative AI systems end to end, from data pipelines to deployed apps. Selected for a competitive Zoom-sponsored fellowship building Zoom Lens.

Education

Arizona State University

Master of Science in Data Science, Analytics and Engineering | GPA: 3.78/4.0

Tempe, AZ

Expected May 2027

Anurag University

Bachelor of Technology in Artificial Intelligence and Machine Learning | GPA: 3.53/4.0

Hyderabad, India

Aug 2021 to May 2025

Technical Skills

Languages: Python, SQL, Java, JavaScript, TypeScript, Rust, C++, C, R, MATLAB

AI & ML: TensorFlow, Scikit-learn, XGBoost, SHAP, NLTK, VADER, OpenCV

APIs & SDKs: Google Gemini API, Zoom Video SDK, Zoom Realtime Media Streams API, REST APIs, GraphQL

Data & Databases: PostgreSQL, MongoDB, Neo4j, SQLite, PySpark

Frameworks & Tools: Flask, Node.js, Express.js, SvelteKit, Streamlit, JWT, Docker, Git, GitHub Actions, ServiceNow

Experience

Zoom Fellow

ASU Next Lab, in partnership with Zoom

Tempe, AZ

Aug 2026 to Present

- Built **Zoom Lens**, a prototype AI assistant inside Zoom meetings that privately describes and explains shared-screen content for individual participants, without interrupting the presenter.
- Implemented Describe, Explain, and contextual follow-up interactions on **Zoom APIs/SDKs** and **generative AI**, with participant-specific request handling that returns each response only to the requester.
- Exploring next steps: continuous screen understanding, transcript-and-screen reasoning, and accessibility support.

Deskside Support Intern

ASU Media and Immersive eXperience (MIX) Center

Mesa, AZ

May 2026 to Aug 2026

- Diagnosed and resolved hardware, software, network, and account issues across classrooms and offices (full-time).
- Tracked incidents and documented resolutions in **ServiceNow**; troubleshooted recurring user and system issues with IT.

Research Assistant

Anurag University

Hyderabad, India

Aug 2023 to May 2025

- Validated structured datasets and academic records across **5+ departments**, keeping data consistent for reporting.
- Reorganized digital filing and retrieval processes, cutting information access time by **20%**.

Projects

CodeAtlas: Rust Change-Impact Analyzer | Rust, tree-sitter, Neo4j, GraphQL, SvelteKit

2026

- Built a static-analysis tool that parses Rust repositories with **tree-sitter** and labels every call site resolved, ambiguous, unresolved, or external instead of guessing, storing the call graph in **Neo4j** for change-impact queries through a **GraphQL** API and SvelteKit UI.
- Added git-diff test selection scored against fault-injection ground truth (**83–100% recall**, **81–89% precision** on small fixture projects); indexed tokio in **3.6s**, with **181 Rust** and **31 web tests** in GitHub CI.

Support Ticket Triage & Resolution-Time Predictor | Python, XGBoost, PySpark, SHAP, Flask, Docker Jul 2026

- Built an ML pipeline on **20,000 synthetic** support tickets comparing Logistic Regression, Random Forest, and **XGBoost**; XGBoost reached **macro-F1 0.705** on priority and cut resolution-time error to **12.2h** from a 24.4h baseline.
- Engineered text features with NLTK and VADER sentiment, added **SHAP** per-prediction explanations, and served the model through a Dockerized Flask REST API (**15–40ms** responses) covered by **25 automated tests**.

Reddit Data ETL Pipeline | PostgreSQL, SQL, Docker, GitHub Actions

Feb 2026

- Built a **5-stage ETL pipeline** (extract, validate, stage, merge, check) that screened 18.9M Reddit lines, rejecting 111,408 invalid, 21,826 duplicate, and 53,793 orphaned records, and loaded **18.71M** clean rows into PostgreSQL.
- Cut full-run time from 584s to **345s** with COPY bulk staging and drop-and-rebuild constraints (merge step 267s to 102s); **49 automated tests** run on every push via GitHub Actions.

StreamBox: Web-Based OTT Platform | Node.js, Express.js, MongoDB, Docker

Sep 2022

- Built a full-stack streaming platform with a **24-endpoint** Express REST API, JWT authentication, and 5 MongoDB collections, using atomic updates and **optimistic locking** to keep concurrent writes consistent.
- Verified concurrency safety with 24 automated race-condition tests and a **100-user load test** (51,633 requests at about 3,440 requests/s, **0** errors, **0** duplicates, **0** incorrect totals).